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(54) **ORTHOGONAL PRINTED CIRCUIT BOARD INTERFACE**

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H01Q 21/00 (2006.01)

H01Q 21/24 (2006.01)

(52) **U.S. Cl.**

CPC **H01Q 21/0025** (2013.01); **H01Q 21/24** (2013.01); **H05K 1/0243** (2013.01); **H05K 2201/10098** (2013.01)

(58) **Field of Classification Search**

CPC .. **H01Q 21/0025**; **H01Q 21/065**; **H01Q 21/24**; **H05K 1/0243**; **H05K 2201/10098**

See application file for complete search history.

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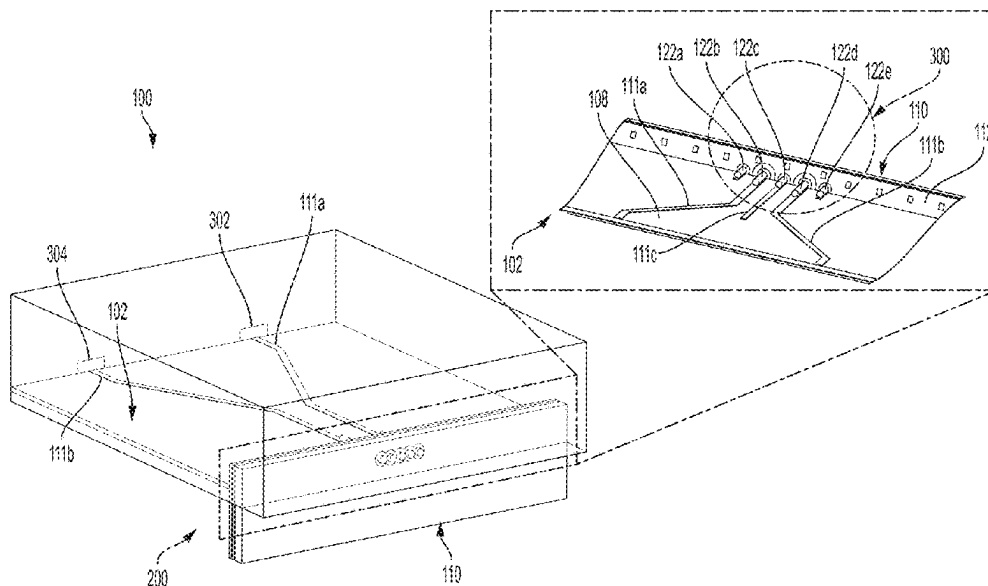
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(57)

ABSTRACT

A dual-polarized PCB array antenna includes one or more PCBs and a radiating antenna array. Each PCB has a PCB mounting surface that extends from a first PCB end to an opposing second PCB end. The radiating antenna array includes one or more radiator substrates. Each radiator substrate has a patch mounting surface that extends from a first substrate end to an opposing second substrate end. The dual-polarized PCB array antenna further includes one or more orthogonal interfaces configured to arrange the patch mounting surface of a given radiator substrates in an orthogonal position with respect to the PCB mounting surface of a given PCB.

7 Claims, 7 Drawing Sheets



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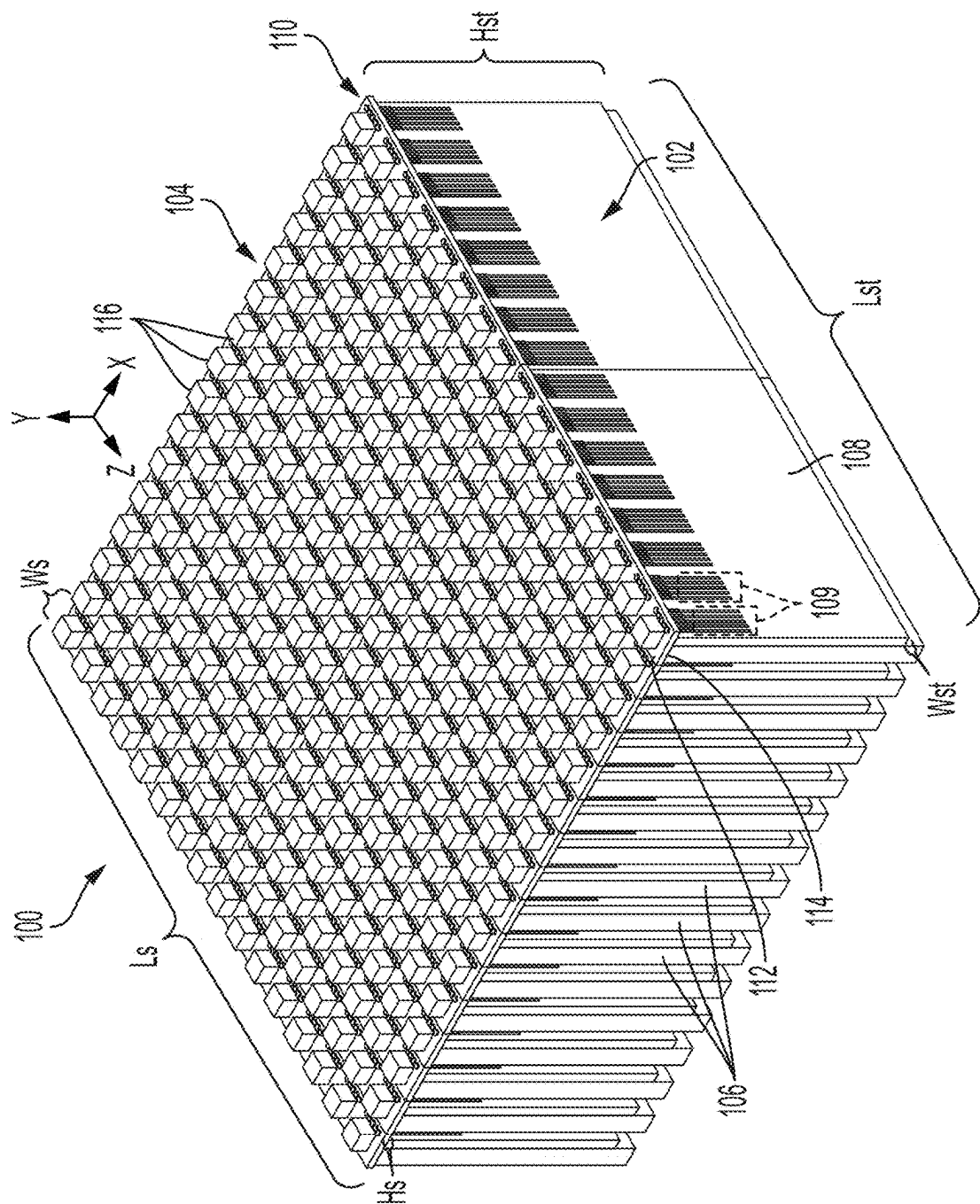
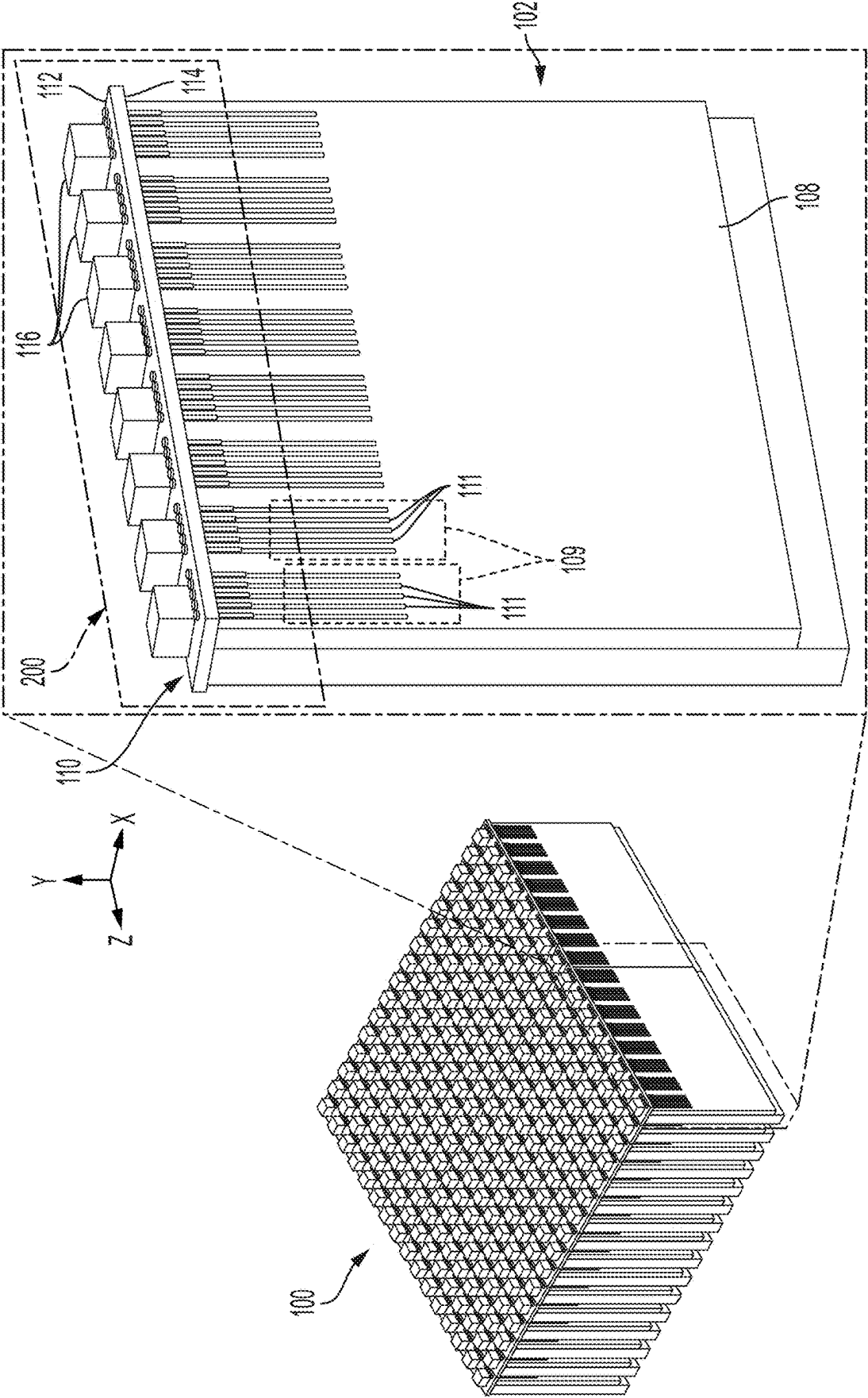


FIG. 1



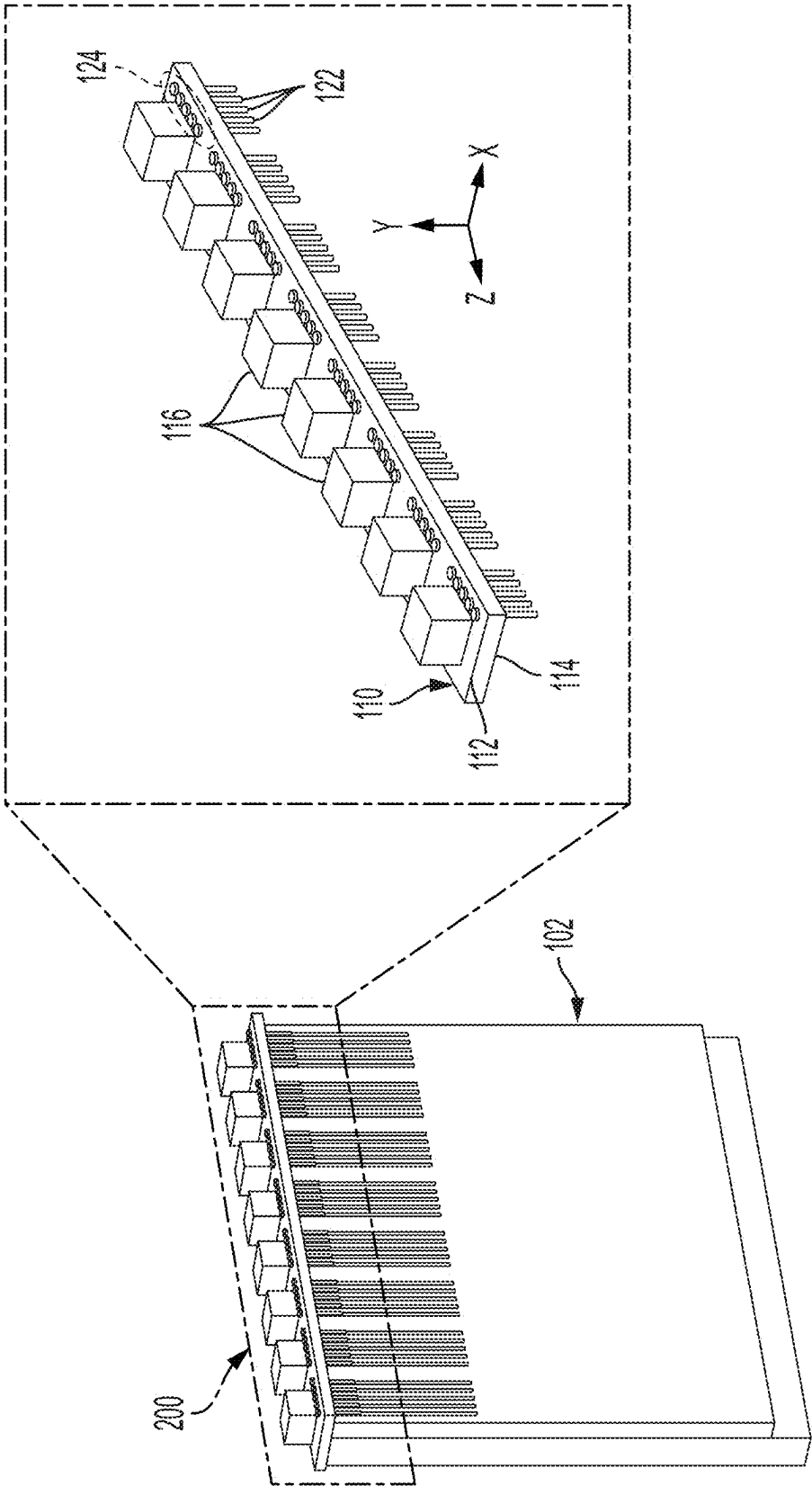


FIG. 3

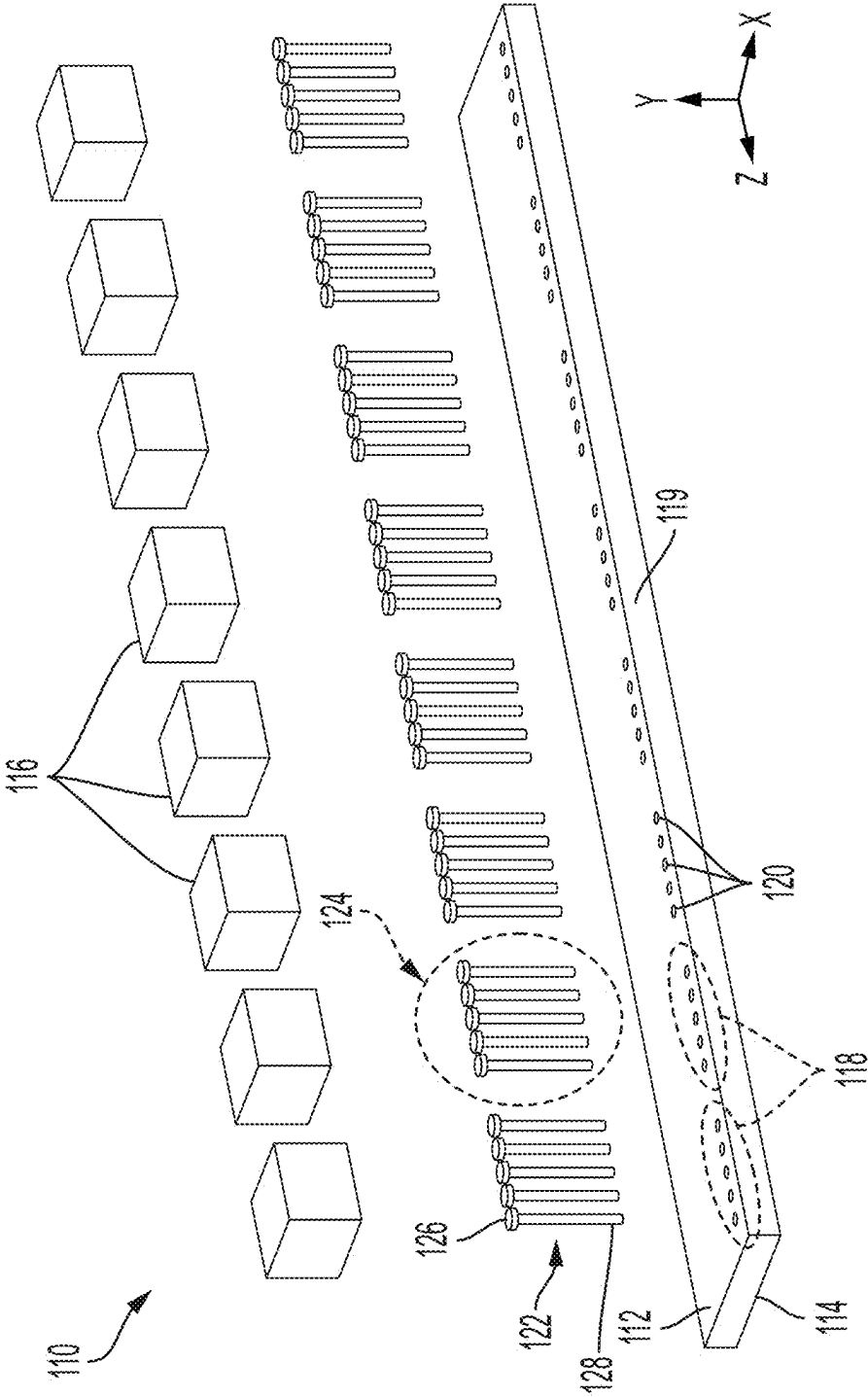


FIG. 4

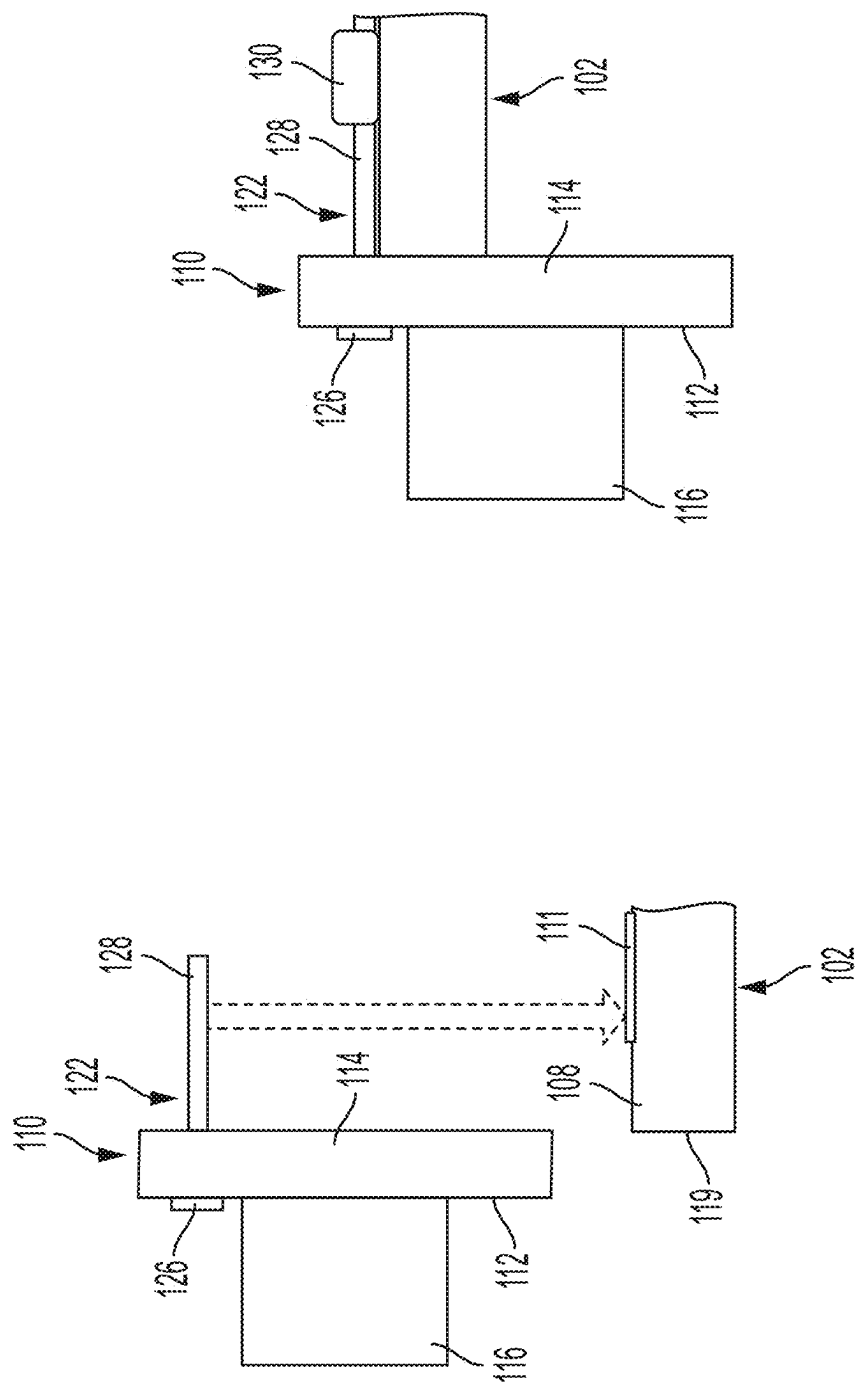


FIG. 5B

FIG. 5A

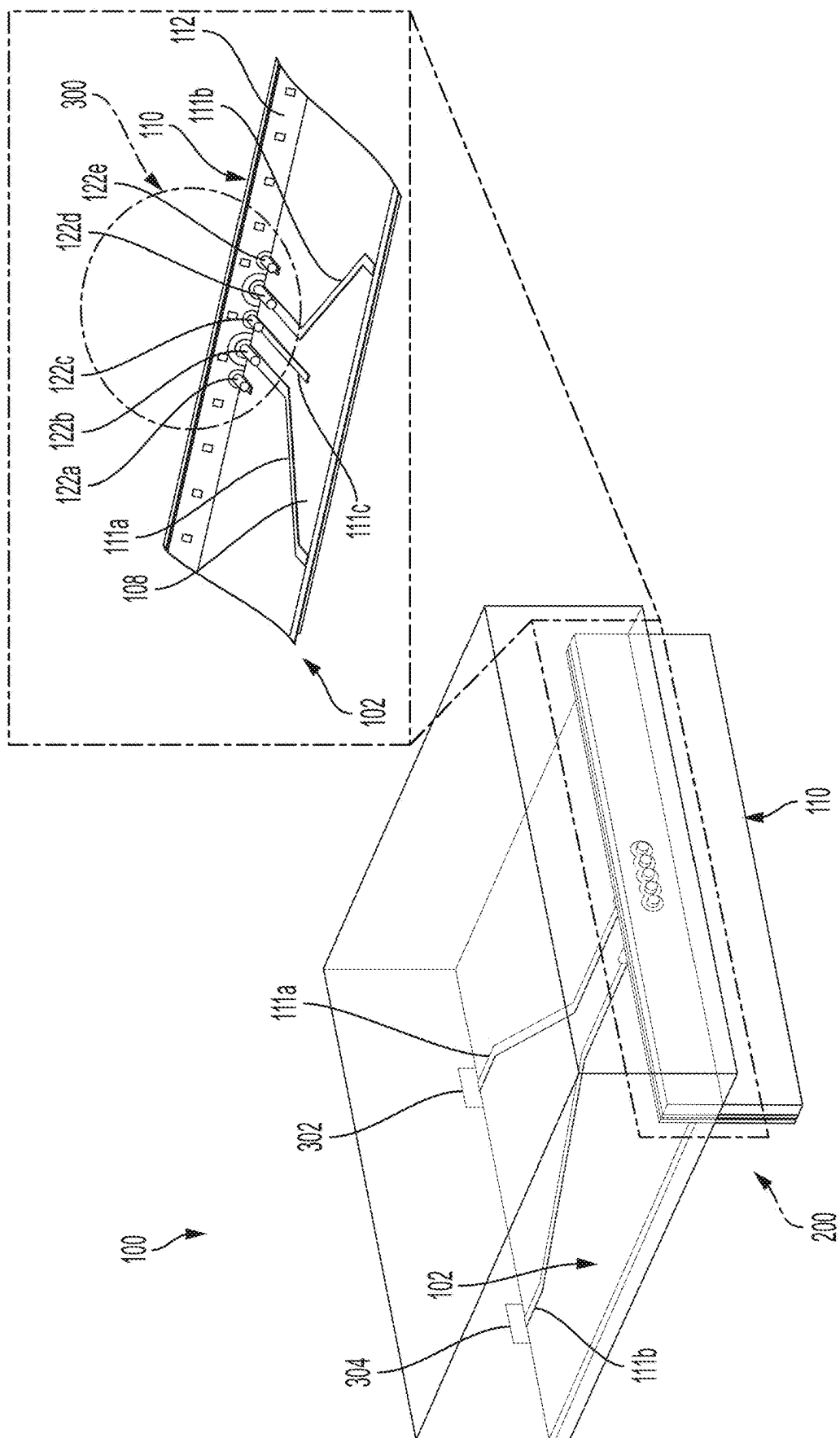
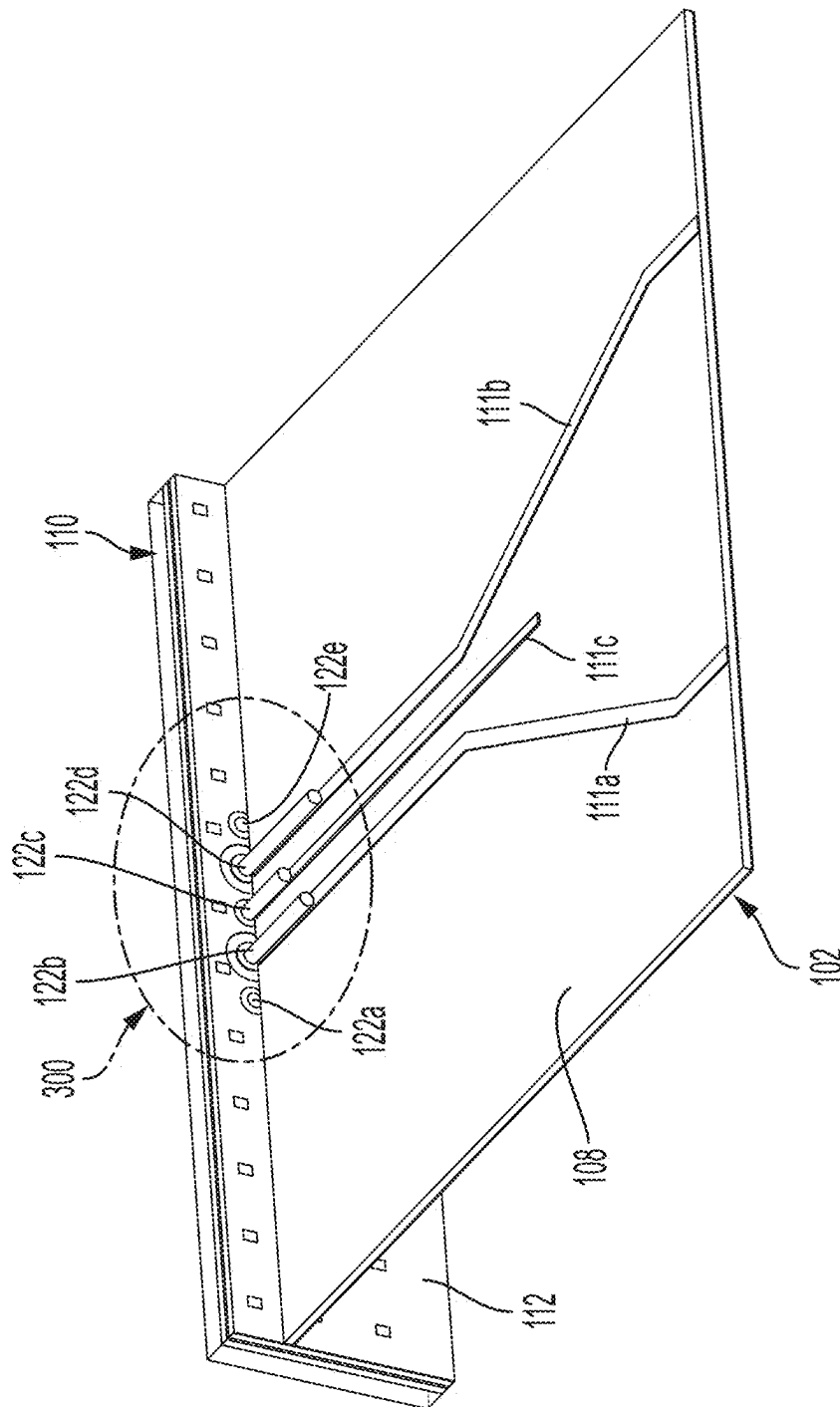


FIG. 6



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ORTHOGONAL PRINTED CIRCUIT BOARD INTERFACE

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a division of U.S. application Ser. No. 17/200,155 filed Mar. 12, 2021, the disclosure of which is incorporated herein by reference in its entirety.

BACKGROUND

The present disclosure relates to antennas and, in particular, to phased array antennas.

Phased array antennas (also referred to as “phased arrays”) are used in communication, radar, and direction-finding systems as well as in other multifunction radio frequency (RF) systems. The phased array antenna typically includes an array of individual radiating antenna elements. The selection of the individual radiating element and arrangement of such elements affect the ability to efficiently transmit and receive RF signals having multiple polarizations.

One example of an architecture used to facilitate a phase array antenna is referred to as a “PCB array.” The PCB array architecture typically implements two separate array packages on opposite surfaces of a relatively long printed circuit boards (PCBs), thus giving rise to the name of a “PCB” array. The large surface area existing across the depth of each individual PCB provides a large area to mount components and transmit/receive modules, while also distributing thermal loads across a large volume.

SUMMARY

According to a non-limiting embodiment, a dual-polarized PCB array antenna is provided. The dual-polarized PCB array antenna comprises a plurality of PCBs and a radiating antenna array. The plurality of PCBs, each PCB extends along a first axis to define a PCB width, a second axis orthogonal to the first axis to define a PCB height, and a third axis orthogonal to the first axis and the second axis to define a PCB length. Each PCB has a PCB mounting surface that extends along the second axis from a first PCB end to an opposing second PCB end and along the third axis from a third PCB end to a fourth PCB end. The radiating antenna array includes a plurality of radiator substrates. Each radiator substrate has a patch mounting surface that extends along the first axis from a first substrate end to an opposing second substrate end and the third axis from a third substrate end to an opposing fourth substrate end. The dual-polarized PCB array antenna further comprises a plurality of orthogonal interfaces configured to arrange the patch mounting surface of the plurality of radiator substrates in an orthogonal position with respect to the PCB mounting surface of the plurality of PCBs.

According to another non-limiting embodiment, an orthogonal printed circuit board (PCB) interface included in a dual-polarized array antenna is provided. The orthogonal PCB interface comprises a PCB and a radiator substrate. The PCB comprises a PCB mounting surface including a first plurality of electrically conductive elements. The radiator substrate comprises a patch mounting surface that includes a second plurality of electrically conductive elements. The second plurality of electrically conductive elements are coupled to the first plurality of electrically conductive elements such that the patch mounting surface of the radiator

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substrate is arranged in an orthogonal position with respect to the PCB mounting surface of the PCB.

According to yet another non-limiting embodiment, a shielded channel interface included in a dual-polarized printed circuit board (PCB) array antenna is provided. The shielded channel interface comprises a plurality of first electrically conductive elements coupled to a PCB mounting surface of a PCB, and a plurality of second electrically conductive elements coupled to a patch mounting surface of a radiator substrate. The plurality of second electrically conductive elements extend perpendicular from the patch mounting surface to establish electrical connections to the plurality of first electrically conductive elements. At least two first connections among the electrical connections are configured to receive a first signal and at least one second connection among the electrical connections is configured to receive a second signal different from the first signal, the at least one second connection interposed between the at least two first connections.

Additional features and advantages are realized through the techniques of the present disclosure. Other embodiments and aspects are described in detail herein and are considered a part of the claimed disclosure. For a better understanding of the disclosure with the advantages and the features, refer to the description and to the drawings.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

For a more complete understanding of this disclosure, reference is now made to the following brief description, taken in connection with the accompanying drawings and detailed description, wherein like reference numerals represent like parts:

FIG. 1 is a perspective view of a dual-polarized PCB array antenna according to a non-limiting embodiment;

FIG. 2 depicts an orthogonal PCB interface included in the dual-polarized PCB array antenna shown in FIG. 1 according to a non-limiting embodiment;

FIG. 3 depicts a radiator PCB included in the orthogonal PCB interface shown in FIG. 2 according to a non-limiting embodiment;

FIG. 4 is a disassembled view of the radiator PCB shown in FIG. 3 according to a non-limiting embodiment;

FIG. 5A depicts a patch antenna included in the radiator PCB shown in FIG. 4 according to a non-limiting embodiment;

FIG. 5B depicts the patch antenna shown in FIG. 5A coupled to a PCB included in the orthogonal radiator PCB interface shown in FIG. 2 according to a non-limiting embodiment;

FIG. 6 depicts a shielded channel interface provided by a dual-polarized PCB array antenna according to a non-limiting embodiment; and

FIG. 7 depicts a shielded channel interface provided by a dual-polarized PCB array antenna according to another non-limiting embodiment.

DETAILED DESCRIPTION

Various non-limiting embodiments described herein provide an orthogonal printed circuit board (PCB) interface for implementation in a dual-polarized PCB array antenna. The orthogonal printed circuit board (PCB) interface overcomes a technology gap that involves using only single PCB, while facilitating efficient dual-polarization via dual-polarized

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patch and stacked patch radiators integrated on a respective radiator substrate that is arranged orthogonally with respect to a mating PCB.

According to at least one non-limiting embodiment, the orthogonal PCB interface provides a high-efficiency connection between a radiator substrate arranged in an orthogonal position with respect to a PCB and completes the needed functionality to exchange signals in a phased array front-end. In addition, the orthogonal PCB interface provides minimal reflection coefficient performance within an operating frequency range compatible with contemporary phased arrays, along with also providing a minimal interference with electronic circuits and/or components (e.g., amplifiers, switches or hybrid circuits) included in the radiator substrate and the PCB. The orthogonal PCB interface also establishes a structural interface that provides a sufficient thermal boundary and exchange medium between the radiator substrate and the PCB. In this manner, the orthogonal PCB interface is capable of withstanding contemporary manufacturing processes and expected stress during phased array operation, transportation, and storage.

With reference now to FIG. 1, a dual-polarized PCB array antenna **100** is illustrated according to a non-limiting embodiment. The dual-polarized PCB array antenna **100** includes a plurality of individual PCBs **102** and a radiating antenna array **104**. The PCBs **102** are arranged side-by-side along a first axis (e.g., an X-axis) and separated from one another by an air gap **106**. Each PCB **102** extends along the first axis (e.g., the X-axis) to define a PCB width (Wst), a second axis (e.g., a Y-axis) orthogonal to the first axis to define a PCB height (Hst), and a third axis (e.g., a Z-axis) orthogonal to the X-axis and the Y-axis to define a PCB length (Lst). The PCBs **102** can be formed from various materials that provide a dielectric constant (k) ranging, for example, from about 2.0 to about 6, a dielectric loss tangent (tan δ) ranging, for example, from about 0.0005 to about 0.03, a coefficient of thermal expansion (CTE) ranging, for example, from about 10 ppm/ $^{\circ}$ C. to about 220 ppm/ $^{\circ}$ C. In one or more embodiments, the PCBs **102** can be fabricated as a printed circuit board (PCB) that includes various circuits, traces, and/or electrical components.

Each PCB **102** has a PCB mounting surface **108** that extends along the Y-axis from a first PCB end (e.g., an upper end) to an opposing second PCB end (e.g., a lower end) and along the Z-axis from a third PCB end (e.g., a left end) to a fourth PCB end (e.g., a right end). The PCB mounting surface **108** includes a plurality of trace clusters **109** arranged side-by-side along the Z-axis. The trace clusters **109** assist in facilitating an interface between a given PCB **102** and the radiating antenna array **104** as described in greater detail below.

The radiating antenna array **104** is disposed on the first PCB end of the plurality of PCBs **102** and is arranged orthogonally with respect to the PCB mounting surface **108**. The radiating antenna array **104** includes a plurality of individual radiator substrates **110** arranged side-by-side along the X-axis. In one or more embodiments, the radiator substrates **110** can be fabricated as a printed circuit board (PCB) that includes various circuits, traces, and/or electrical component.

Each radiator substrate **110** is disposed on the first end of a respective PCB **102**, and extends along the first axis (e.g., the X-axis) to define a substrate width (Ws), a second axis (e.g., a Y-axis) orthogonal to the first axis to define a substrate height (Hs), and a third axis (e.g., a Z-axis) orthogonal to the X-axis and the Y-axis to define a substrate length (Ls). The radiator substrates **110** can be formed from

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various materials that provide a dielectric constant (k) ranging, for example, from about 1 to about 4, a dielectric loss tangent (tan δ) ranging, for example, from about 0.0005 to about 0.03, a coefficient of thermal expansion (CTE) ranging, for example, from about 10 ppm/ $^{\circ}$ C. to about 220 ppm/ $^{\circ}$ C. In one or more embodiments, the radiator substrates **110** can be fabricated as a printed circuit board (PCB) that includes various circuits, traces, and/or electrical components.

Each radiator substrate **110** has a patch mounting surface **112** and an opposing contact surface **114** that extend along the X-axis from a first substrate end to an opposing second substrate end and the Z-axis from a third substrate end to an opposing fourth substrate end. Accordingly, the contact surface **114** is disposed directly against the first end of a respective PCB **102** such that the patch mounting surface **112** is arranged orthogonally with respect to the PCB mounting surface **108**. In one or more non-limiting embodiments, the first and second ends of the radiator substrates **110** directly contact one another.

Each radiator substrate **110** includes a plurality of radiating antenna elements **116** disposed on the patch mounting surface **112** and arranged side-by-side along the Z-axis. Although the radiating antenna elements **116** are described as patch antennas **116** going forward, other types of radiating antenna elements **116** can be implemented without departing from the scope of the invention.

Referring to FIG. 2, a PCB **102** and a radiator substrate **110** included in the dual-polarized PCB array antenna **100** are illustrated in greater detail. The PCB **102** is illustrated with trace clusters **109** that include a plurality of electrically conductive traces **111** formed on the PCB mounting surface **108**. The traces **111** include an electrically conductive material (e.g., copper) and can be formed on the PCB mounting surface **108** using various fabrication processes such as, for example, printed circuit board (PCB) etching processes, additive manufacturing, (e.g., three-dimensional printing), etc. Although five traces **111** are shown, it should be appreciated that the trace clusters **109** can include more or less traces **111** without departing from the scope of the invention.

Still referring to FIG. 2 with additional reference to FIGS. 3 and 4, the radiator substrate **110** has a plurality of through-hole clusters **118** arranged side-by-side along the Z-axis (see FIG. 4). Each through-hole cluster **118** is interposed between a given patch antenna **116** and the first end **119** of the patch mounting surface **112**. Each through-hole cluster **118** includes a plurality of through-holes **120** formed in the radiator substrate **110** and extending completely there through from the patch mounting surface **112** to the contact surface **114**. Each through-hole **120** is aligned with a respective trace **111** along the Y-axis (shown e.g., in FIG. 2).

The through-holes **120** are configured to receive electrically conductive pins **122** to provide a plurality of pin clusters **124**, which are arranged side-by-side along the Z-axis. The pins **122** are formed from an electrically conductive material such as copper, for example, so as to serve as terminals. Each electrically conductive pin **122** extends from a mounting end **126** to an opposing contact end **128** and is disposed in a respective through-hole **120** (see FIG. 3). Accordingly, the mounting end **126** abuts against the patch mounting surface **112** and the contact end **128** contacts an aligned trace **111** as shown in FIG. 2. In one or more non-limiting embodiments, the radiator substrate **110** can include one or more substrate traces (not shown) connecting a given patch antenna **116** to the pins **122** included in an adjacent pin cluster **124**. In this manner, signals (e.g., RF signals) applied to the traces **111** of a given trace cluster **109**

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on the PCB 102 can be deliver to the patch antenna 116 on the radiator substrate 110 via the respectively aligned pins 122. Although five pins 122 are shown, more or less pins 122 can be included based on the number of traces 111 formed on the PCB mounting surface 108.

Referring again to FIG. 2, the assembly of the PCB 102, the traces 111, the radiator substrate 110, and the pins 122 establishes an orthogonal PCB interface 200. As described herein, the orthogonal PCB interface 200 arranges the patch mounting surface 112 of the radiator substrate 110 in an orthogonal position with respect to the PCB mounting surface 108 of the PCB 102. This orthogonal arrangement allows for integrating dual-polarized patch antennas or stacked patch radiators with the PCBs 102 to enable a dual-polarized PCB array antenna 100. In addition, the electrical connection established by the traces 111 and the pins 122 can facilitate a shielded channel interface between the PCB 102 and the patch antennas 116 as described in greater detail below.

The connection between the pins 122 and the traces 111 also facilitates dual-polarization RF signal transmission and/or reception without the need to implement large, bulky right-angle connectors. FIGS. 5A and 5B illustrate one example of establishing the pin/trace connection. The pins 122 can be aligned above the traces 111 using, for example a pick-and-place device as understood by those of ordinary skill in the art. Once aligned, the radiator substrate 110 can be placed (as indicated by the dashed arrow) against the PCB 102 such that the contact surface 114 abuts against the first end 119 of the PCB 102 and the pins 122 contact the traces 111 (see FIG. 5A). An electrically conductive filler 130 such as solder, for example, can then be deposited on the PCB mounting surface 108 so as to form an electrically conductive node between the contact end 128 of the pins 122 and the traces 111 (see FIG. 5B). Although the electrically conductive filler 130 is shown as covering a portion of the pin contact end 128, it should be appreciated that the electrically conductive filler 130 can be deposited to cover the entire contact end 128.

Turning to FIG. 6, a shielded channel interface 300 is provided by a dual-polarized PCB array antenna 100 is illustrated according to a non-limiting embodiment. As described herein, the dual-polarized PCB array antenna 100 includes a one or more PCBs 102 arranged orthogonally with respect to a radiator substrates 110 via an orthogonal PCB interface 200. Although a single PCB 102 and a single radiator substrate 110 are illustrated, it should be appreciated that additional PCBs 102 and additional substrates 110 can be implemented without departing from the scope of the invention.

According to a non-limiting embodiment shown in FIG. 6, the PCB 102 includes a first plurality of electrically conductive elements 111a, 111b and 111c. In one or more non-limiting embodiments, the first plurality of electrically conductive elements include a first electrically conductive trace 111a, a second electrically conductive trace 111b, and a third electrically conductive trace 111c. In at least one non-limiting embodiment, one or more of the electrically conductive traces 111a-111c are formed on the PCB mounting surface 108 and extend from the first PCB end to the second PCB end. The first trace 111a can be connected to a first port 302, the trace 111b can be connected to a second port 304, and the third electrically conductive trace 111c can be connected to a ground plane using a via (not shown) embedded in the PCB 102. Although the first and second ports 302 and 304 is shown located at the second PCB end,

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the locations of the first and second ports 302 and 304 are not limited thereto and may be implemented at different locations.

The radiator substrate 110 includes a second plurality of electrically conductive elements 122a, 122b, 122c, 122d, and 122e (collectively referred to as 122a-122e). In at least one non-limiting embodiment, the electrically conductive elements 122a-122e can be formed as electrically conductive pins as described herein, which serve as a plurality of terminals to facilitate the shielded channel interface 300. In at least one non-limiting embodiment, the electrically conductive elements 122a-122e extend perpendicular with respect to the patch mounting surface 112 of the radiator substrate 110. Accordingly, a second electrically conductive element (e.g., a pin 122a-122e) can contact a respective first electrically conductive element (e.g., a trace 111a-111c).

As shown in FIG. 6, for example, terminal 122b can be connected to the first trace 111a while terminal 122d can be connected to the second trace 111b. Terminals 122a, 122c and 122e can be connected to vias (not shown), which are embedded in the PCB 102, and are shorted to a ground plane. Accordingly, a ground-signal-ground-signal-ground (GSGSG) shielded channel interface 300 is established that allows for maintaining ground potentials between both the PCB 102 and the radiator substrate 110 throughout the system while RF signals are applied to terminals 122b and 122d and delivered to the first and second terminals 111a and 111b, respectively. It should be appreciated that the terminals 122a-122e are not limited to a GSGSG configuration, and that other terminal configurations capable of establishing a shielded channel interface 300 can be employed without departing from the scope of the present disclosure.

According to a non-limiting embodiment shown in FIG. 7, a shielded channel interface 300 can be established by utilizing the inner terminals 122b, 122c and 122d without utilizing the outer-most terminals 122a and 122e. In this example, terminal 122c can serve as a ground terminal that is interposed (i.e., sandwiched) between first and second RF terminals 122b and 122d. Accordingly, an RF signal-ground-signal (SGS) shielded channel interface 3000 is established that facilitates a signal interface between the PCB 102 and the radiator substrate 110. Although a SGS shielded channel interface 300 is described, the configuration is not limited thereto. For example, a ground-signal-ground (GSG) shielded channel interface 300 can be established without departing from the scope of the present disclosure by utilizing terminal 122c as an RF terminal that is interposed (i.e., sandwiched) between first and second ground terminals 122b and 122d.

As described herein, various non-limiting embodiments provide an orthogonal interface for implementation in a dual-polarized PCB array antenna. The orthogonal interface overcomes the need to implement twin PCBs, while facilitating efficient dual-polarization via dual-polarized patch and stacked patch radiators integrated on one or more PCBs.

The corresponding structures, materials, acts, and equivalents of all means or step plus function elements in the claims below are intended to include any structure, material, or act for performing the function in combination with other claimed elements as specifically claimed. The description of the present invention has been presented for purposes of illustration and description, but is not intended to be exhaustive or limited to the invention in the form disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the invention. The embodiments were chosen and described in order to best explain the principles of the

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invention and the practical application, and to enable others of ordinary skill in the art to understand the invention for various embodiments with various modifications as are suited to the particular use contemplated.

While the preferred embodiments to the invention have been described, it will be understood that those skilled in the art, both now and in the future, may make various improvements and enhancements which fall within the scope of the claims which follow. These claims should be construed to maintain the proper protection for the invention first described.

What is claimed is:

1. A shielded channel interface included in a dual-polarized printed circuit board (PCB) array antenna, the shielded channel interface comprising:

a plurality of first electrically conductive elements coupled to a PCB mounting surface of a PCB;

a plurality of second electrically conductive elements coupled to a patch mounting surface of a radiator substrate, the plurality of second electrically conductive elements extending perpendicular from the patch mounting surface to establish electrical connections to the plurality of first electrically conductive elements, wherein at least two first connections among the electrical connections are configured to receive a first signal and at least one second connection among the electrical

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connections is configured to receive a second signal different from the first signal, the at least one second connection interposed between the at least two first connections.

2. The shielded channel interface of claim 1, wherein the electrical connections establish a ground-signal-ground configuration.

3. The shielded channel interface of claim 1, wherein the electrical connections establish a signal-ground-signal configuration.

4. The shielded channel interface of claim 1, wherein the plurality of first electrically conductive elements include at least one electrically conductive trace formed on the PCB mounting surface.

5. The shielded channel interface of claim 4, wherein each of the at least one electrically conductive trace includes a first end in signal communication with a port and a second end in signal communication with a ground plane.

6. The shielded channel interface of claim 5, wherein the plurality of second electrically conductive elements include at least one electrically conductive pin.

7. The shielded channel interface of claim 6, wherein the at least one electrically conductive pin is connected to the at least one electrically conductive trace.

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